Numerical Validation of Structural Selection: Evidence Beyond Energy-Based Ranking in a Minimal ϕ^4 Model

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Abstract

Structural Selection Theory distinguishes between *dynamical admissibility*, encoded by a base action S_0, and *physical realisation*, encoded by an additional structural functional $\mathcal{F}_{\text{struct}}$. The companion model paper formulated this idea explicitly for a $1+1$ dimensional ϕ^4 field theory by defining five operational diagnostics (H, B, S, V, R) and a sector-sensitive structural score $\mathcal{F}_{\text{struct}}[\phi; q]$.

The present paper reports the first numerical benchmark study of that model on a static lattice. Three protocols are evaluated: vacuum discrimination in the trivial topological sector, kink discrimination in the unit sector, and a direct comparison between structural ranking and static-energy ranking in a distortion ensemble. The robustness sector is implemented as a one-sided instability penalty, so that unstable Hessian directions are penalised while stable configurations are not artificially rewarded for additional stiffness.

The benchmark yields three main results. First, the true vacua are cleanly separated from the unstable homogeneous saddle, with a structural margin $\Delta_{\text{vac}} = 2.583852$. Second, the relaxed discrete kink outperforms all tested sector-matched distortions, giving $p_K = 1.000$. Third, structural ranking matches energy ranking at the very closest percentiles and improves upon it at the 20% level, where the mean distance to the discrete kink decreases from 0.012128 (energy ranking) to 0.010363 (structural ranking).

These results do not yet establish a universal selection law. They do show that the proposed structural layer is calculable, falsifiable, and already adds nontrivial discriminatory power in a minimal nonlinear field-theory setting.

1 Introduction

Many nonlinear field theories admit multiple exact or near-exact configurations with sharply different physical character: stable vacua, unstable saddles, topological solitons, rough deformations, and metastable intermediates. Ordinary variational dynamics explains which configurations are admissible, but it does not by itself provide a dedicated scalar functional for ranking the relative structural quality of those admissible or nearly admissible states.

Structural Selection Theory (SST) was introduced to formalise that missing layer. Its central idea is the separation

$$S_0 \;\Rightarrow\; \text{admissibility}, \qquad \mathcal{F}_{\text{struct}} \;\Rightarrow\; \text{realisation}.$$

In the accompanying construction for the ϕ^4 model, the structural functional is built from five explicit diagnostics:

- H: equation consistency,
- B: conservation consistency,
- S: structural activity,
- V: nonlocal coherence,

- R: robustness against unstable fluctuations.

The present paper asks the first strictly numerical question:

> *Does the structural functional $\mathcal{F}_{\text{struct}}$ produce clean, reproducible, and nontrivial rankings in a minimal nonlinear benchmark?*

To answer this, we test three concrete protocols on a static lattice:

1. discrimination of the true vacua from the unstable homogeneous saddle,
2. discrimination of the discrete kink from sector-matched distortions,
3. comparison of structural ranking against ordinary static-energy ranking.

The goal is deliberately modest. This is not yet a claim of universality. It is the first serious check that the proposed structural diagnostics can be turned into an operational numerical pipeline.

2 Model and Structural Functional

We consider the standard $1+1$ dimensional ϕ^4 model on a flat background:

$$S_0[\phi] = \int dt\, dx \left[\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{\lambda_\phi}{4}\left(\phi^2 - v^2\right)^2\right].$$

The static continuum equation in the dimensionless units used below becomes

$$-\phi'' + \frac{1}{2}\phi\left(\phi^2 - 1\right) = 0.$$

The relevant exact configurations are the two true vacua $\phi_\pm = \pm 1$, the unstable homogeneous saddle $\phi_0 = 0$, and the unit-charge kink sector $Q_{\text{top}} = 1$.

The structural functional used in the numerical study is

$$\begin{aligned}\mathcal{F}_{\text{struct}}[\phi; q] = {} & \alpha_H\left(1 - H[\phi]\right) + \alpha_B\left(1 - B[\phi]\right) + \alpha_S S[\phi] \\ & + \alpha_V\left(1 - V[\phi]\right) + \alpha_R\left(1 - R[\phi]\right) + \alpha_Q\left(Q_{\text{top}}[\phi] - q\right)^2. \end{aligned} \tag{1}$$

The decisive refinement relative to the earlier exploratory version is the robustness term:

$$R[\phi] = \exp\left[-\gamma_R\,\frac{\max(0, -\lambda_*[\phi])}{m^2}\right].$$

Here $\lambda_*[\phi]$ denotes the relevant lowest Hessian eigenvalue after projecting out symmetry zero modes when required. This is a *one-sided* instability penalty: unstable directions are suppressed, whereas stable configurations saturate at $R = 1$ rather than being rewarded simply for larger positive curvature. Numerically, this turns out to be essential for obtaining a clean vacuum benchmark.

3 Static Lattice Protocol

3.1 Discretisation

The static benchmarks are performed on the interval $[-L, L]$ with

$$L = 20, \qquad N_x = 801, \qquad a = \frac{2L}{N_x - 1} = 0.05,$$

and dimensionless parameters

$$v = 1, \qquad m = 1, \qquad \lambda_\phi = \frac{1}{2}.$$

For lattice sites $x_j = -L + (j-1)a$, the discrete residual is

$$r_j[\phi] = -\frac{\phi_{j+1} - 2\phi_j + \phi_{j-1}}{a^2} + \frac{1}{2}\phi_j(\phi_j^2 - 1).$$

The lattice diagnostics follow the companion model paper directly:

- H_{lat} from the squared residual,
- B_{lat} from the squared conservation residual,
- S_{lat} from the static gradient content,
- V_{lat} from a Gaussian nonlocal coherence kernel,
- R_{lat} from the one-sided Hessian instability penalty.

The static Hessian is the tridiagonal matrix

$$(\mathbf{L}_\phi)_{jk} = \left[\frac{2}{a^2} + \frac{1}{2}(3\phi_j^2 - 1)\right]\delta_{jk} - \frac{1}{a^2}\delta_{j,k+1} - \frac{1}{a^2}\delta_{j,k-1}.$$

In the $q = 1$ sector, the near-translation mode is identified by maximal overlap with the discrete derivative of the profile and is removed before defining λ_*.

3.2 Benchmark parameters

For the first static benchmark, the following weights are used:

$$(\alpha_H, \alpha_B, \alpha_V, \alpha_R, \alpha_Q) = (4, 1, 1, 3, 10), \qquad \alpha_S = 0,$$

with

$$\ell_S = \ell_V = 1, \qquad \gamma_R = 4.$$

Setting $\alpha_S = 0$ is intentional. In a static benchmark, the activity sector is not needed for the first discrimination tests, and turning it off prevents an artificial penalty against nontrivial but static coherent structures such as the kink.

3.3 Three benchmark protocols

The numerical study evaluates:

1. **Protocol A:** vacuum discrimination in the trivial sector $q = 0$.
2. **Protocol B:** kink discrimination in the unit sector $q = 1$.
3. **Protocol C:** comparison of structural ranking and static-energy ranking inside the same $q = 1$ distortion ensemble.

The reference kink is not inserted analytically but obtained by numerical relaxation with fixed boundary values $\phi_1 = -1$ and $\phi_{N_x} = +1$. The relaxation converged after 19 899 steps with maximal update 7.997×10^{-14} and maximal residual 9.996×10^{-11}, so the reference profile is numerically on-shell to high precision.

4 Numerical Results

4.1 Protocol A: vacuum discrimination

Table 1 summarises the exact-state benchmark in the trivial sector. The two true vacua receive vanishing structural score, whereas the unstable homogeneous saddle is heavily penalised solely through the robustness sector.

Table 1: Exact-state benchmark in the trivial sector.

State	λ_*	R	$\mathcal{F}_{\text{struct}}$	E_{stat}
$\phi_+ = +1$	1.006168	1.000000	0.000000	0.000000
$\phi_- = -1$	1.006168	1.000000	0.000000	0.000000
$\phi_0 = 0$	-0.493832	0.138716	2.583852	4.993750

The resulting vacuum margin is

$$\Delta_{\text{vac}} = \mathcal{F}_{\text{struct}}[\phi_0; 0] - \mathcal{F}_{\text{struct}}[\phi_\pm; 0] = 2.583852.$$

This clean separation is the first key result: the structural functional does not merely reproduce on-shellness, since all three exact homogeneous states are equation-consistent, yet only the unstable one is structurally penalised.

Beyond the exact states, the perturbed-vacuum family

$$\phi_\pm^{(\eta)}(x_j) = \pm 1 + \eta \cos\left(\frac{\pi x_j}{2L}\right)$$

was evaluated for $\eta \in \{0.05, 0.10, \ldots, 0.50\}$. The expected ordering

$$\mathcal{F}_{\text{struct}}[\phi_\pm; 0] < \mathcal{F}_{\text{struct}}[\phi_\pm^{(\eta)}; 0] < \mathcal{F}_{\text{struct}}[\phi_0; 0]$$

holds throughout the tested range, including the moderate benchmark window $\eta \leq 0.30$. Figure 1 shows the monotone increase of $\mathcal{F}_{\text{struct}}$ under growing vacuum distortion.

4.2 Protocol B: kink discrimination

In the unit topological sector, the relaxed discrete kink attains

$$\mathcal{F}_{\text{struct}}[\phi_K^{\text{lat}}; 1] = 0.000398, \qquad E_{\text{stat}}[\phi_K^{\text{lat}}] = 0.666611.$$

The corresponding diagnostic values are

$$H = 1, \quad B = 1, \quad S = 0.016392, \quad V = 0.999602, \quad R = 1.$$

A distortion ensemble was then generated from

$$\phi_j^{(\varepsilon,k)} = \phi_{K,j}^{\text{lat}} + \varepsilon\, e^{-x_j^2/\sigma^2} \sin\left(\frac{k\pi(x_j + L)}{2L}\right),$$

with $\sigma = 5$, $k \in \{1, 2, 3\}$, and $\varepsilon \in \{0.02, 0.04, \ldots, 0.30\}$.

The main success statistic is

$$p_K = \frac{\#\left\{(\varepsilon, k) : \mathcal{F}_{\text{struct}}[\phi^{(\varepsilon,k)}; 1] > \mathcal{F}_{\text{struct}}[\phi_K^{\text{lat}}; 1]\right\}}{\#\left\{(\varepsilon, k)\right\}}.$$

Numerically,

$$p_K = 1.000,$$

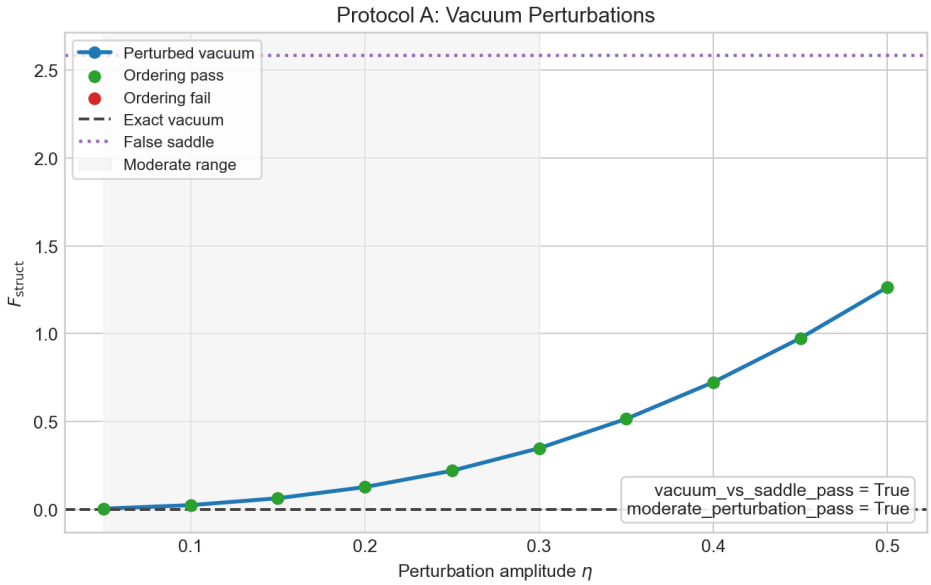

Figure 1: Protocol A: structural score of perturbed vacua as a function of the distortion amplitude η. The exact vacuum remains the minimum, while the unstable saddle lies far above the entire perturbation family.

so the reference kink outperforms every tested distortion. The first and last members of the ensemble illustrate the spread:

$$\mathcal{F}_{\text{struct}} = 0.000462 \quad (\varepsilon = 0.02,\ k = 1),$$

$$\mathcal{F}_{\text{struct}} = 0.008993 \quad (\varepsilon = 0.30,\ k = 3).$$

Figure 2 shows the full result. Across all three distortion families, the discrete kink remains the lowest structural state in the sector.

4.3 Protocol C: structural versus energy ranking

The most demanding benchmark is not simple discrimination but ranking quality. The same $q = 1$ distortion ensemble was therefore ranked in two ways:

1. by structural score $\mathcal{F}_{\text{struct}}$,
2. by ordinary static energy E_{stat}.

For each percentile $p \in \{5\%, 10\%, 20\%\}$, the mean distance to the relaxed discrete kink was computed among the lowest-ranked fraction of the ensemble. The result is shown in Table 2 and Figure 3.

Table 2: Ranking comparison between structural score and static energy.

Percentile	$\overline{d}_K^{(F)}$	$\overline{d}_K^{(E)}$	Structural better?
5%	0.005835	0.005835	No
10%	0.007397	0.007397	No
20%	0.010363	0.012128	Yes

Two points are important here. First, structural ranking does *not* magically dominate at every scale; at the two smallest percentiles it coincides with ordinary energy ranking. Second, at

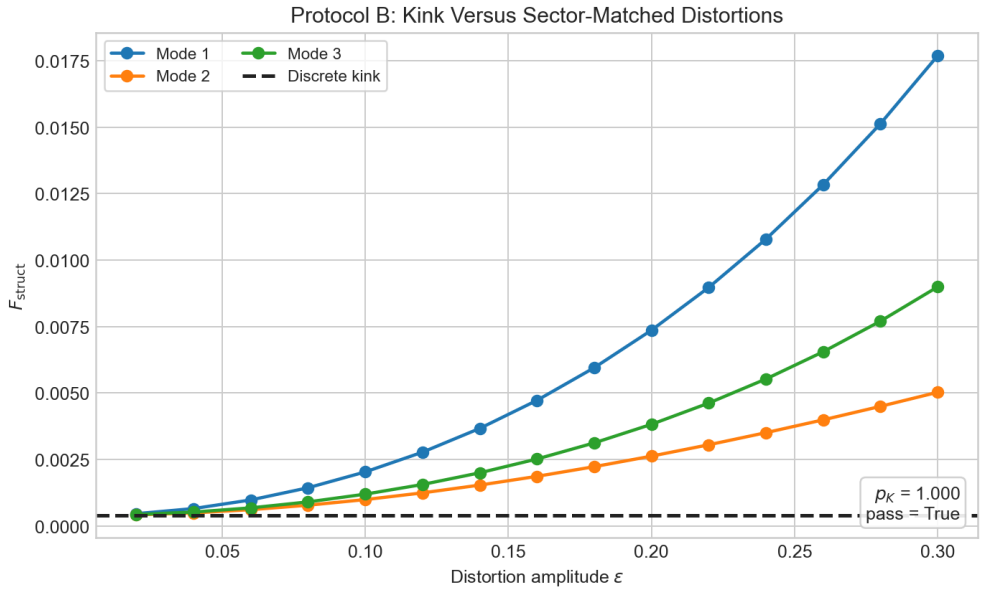

Figure 2: Protocol B: structural score of the relaxed discrete kink versus sector-matched distortions. The reference kink remains below all tested deformations, giving $p_K = 1.000$.

the 20% level it becomes genuinely better:

$$\bar{d}_K^{(F)}(20\%) = 0.010363 < 0.012128 = \bar{d}_K^{(E)}(20\%).$$

This is the first indication that the structural functional adds information beyond ordinary static energy once the ensemble is broad enough that coherence and instability begin to matter jointly.

5 Interpretation

The results support three substantive conclusions.

1. **The structural layer is not vacuous.** Protocol A shows that $\mathcal{F}_{\text{struct}}$ distinguishes stable and unstable exact solutions even when equation consistency alone cannot.
2. **Sectorwise structural selection is numerically meaningful.** Protocol B shows that in a nontrivial topological sector the relaxed discrete kink is not merely on-shell; it is also structurally preferred over a broad distortion ensemble.
3. **Structural ranking is not reducible to energy ranking.** Protocol C shows exact agreement at the sharpest percentiles and a real gain at the 20% level. This is precisely the kind of result one would expect from a selection functional that is related to, but not identical with, static energy.

At the same time, the limitations are equally important. The benchmark remains a one-field, static, flat-space test. In this regime, the balance sector is still partly slaved to the equation residual, and the activity sector was turned off for the first clean discrimination study. Stronger independence between the diagnostics should emerge only in genuinely time-dependent, multi-field, or geometric settings.

6 Reproducibility

The numerical pipeline used for this paper is fully reproducible within the workspace and is prepared for inclusion in the public repository:

https://github.com/Chris4081/structural-selection-principle

From the repository root, the full benchmark can be regenerated by executing:

```
python3 structural_selection_phi4_protocol.py
-json-output structural_selection_phi4_results.json -pretty-json
python3 structural_selection_phi4_plots.py
-input structural_selection_phi4_results.json
-output-dir structural_selection_phi4_plots
```

This produces the numerical result file `structural_selection_phi4_results.json` and the figure directory `structural_selection_phi4_plots/`.

In the repository layout used for dissemination, the numerical scripts are placed alongside the existing analysis code, while the compiled paper is stored in the `documentation/` directory together with the rest of the paper series.

This structure is intentional. The paper is not only a conceptual statement but a small closed reproducibility loop:

$$\text{model} \rightarrow \text{protocol} \rightarrow \text{data} \rightarrow \text{figures}.$$

7 Conclusion

This paper has taken the first numerical step from structural-selection theory to operational benchmark evidence. In a minimal ϕ^4 model, the structural functional $\mathcal{F}_{\text{struct}}$:

- cleanly separates true vacua from the unstable homogeneous saddle,
- ranks the discrete kink above all tested sector-matched distortions,
- and provides a measurable ranking advantage over static energy at the 20% percentile level in the distortion ensemble.

These results are still local in scope, but they already clear an important threshold: the structural layer is no longer philosophical language alone. It has been instantiated as a concrete functional, tested against explicit benchmarks, and shown to produce falsifiable numerical structure.

The next natural step is to move beyond the static one-field case and test whether the same architecture survives in genuinely dynamical, multi-field, or geometric systems.

References

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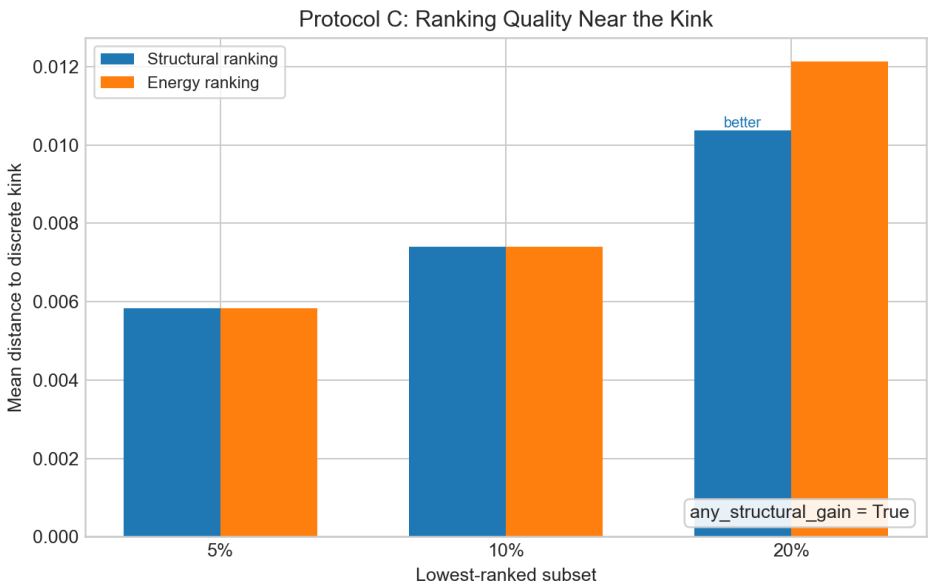
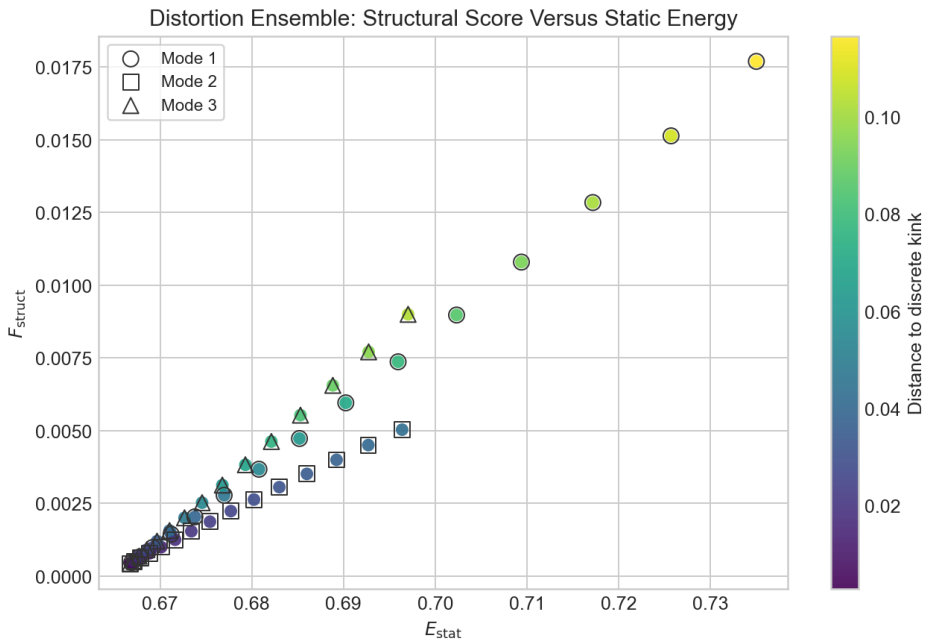


(a) Mean distance to the discrete kink under structural and energy ranking.



(b) Distortion ensemble in the plane $(E_{\text{stat}}, \mathcal{F}_{\text{struct}})$, coloured by distance to the discrete kink.

Figure 3: Protocol C: ranking comparison and ensemble structure. Structural ranking matches energy ranking at the very closest percentiles and improves upon it at the 20% level.